

Code:

System running loop flag

keep_running = "yes"

while keep_running == "yes":

print("\n--- Start a New Food Delivery Order ---")

1. Collect inputs using clean, descriptive variable names

order_amt = float(input("Enter order amount (INR): "))

distance = float(input("Enter delivery distance (km): "))

cust_type = input("Enter customer type (VIP/Regular): ").strip()

cust_rating = float(input("Enter customer rating (1.0 - 5.0): "))

rest_rating = float(input("Enter restaurant rating (1.0 - 5.0): "))

prep_time = int(input("Enter preparation time (minutes): "))

pay_method = input("Enter payment method (Prepaid/COD): ").strip()

weather = input("Enter weather condition (Clear/Rainy/Stormy): ").strip()

demand = input("Enter demand level (Low/Normal/High): ").strip()

peak_input = input("Is it a peak hour? (Yes/No): ").strip()

cancellations = int(input("Enter number of previous cancellations: "))

Normalize text fields to avoid match failures due to letter-case

cust_type = cust_type.upper()

pay_method = pay_method.upper()

weather = weather.capitalize()

demand = demand.capitalize()

peak_input = peak_input.lower()

Convert peak hour input to explicit boolean flag

if peak_input == "yes":

is_peak_hour = True

else:

```
is_peak_hour = False
```

2. Determine Restaurant Status

```
if rest_rating < 3.0:
```

```
    rest_status = "Unreliable"
```

```
elif rest_rating < 4.0 or prep_time > 45:
```

```
    rest_status = "Under Review"
```

```
else:
```

```
    rest_status = "Reliable"
```

3. Determine Cancellation Risk

```
if cancellations > 3 and cust_rating < 3.0:
```

```
    cancel_risk = "High"
```

```
elif cancellations > 1 or cust_rating < 4.0:
```

```
    cancel_risk = "Medium"
```

```
else:
```

```
    cancel_risk = "Low"
```

4. Determine Final Order Status and Manual Review Status

```
if rest_status == "Unreliable" or (cancel_risk == "High" and pay_method == "COD"):
```

```
    order_status = "Rejected"
```

```
    review_status = "Not Required"
```

```
elif rest_status == "Under Review" or order_amt > 5000 or weather == "Stormy":
```

```
    order_status = "Manual Review"
```

```
    review_status = "Pending"
```

```
else:
```

```
    order_status = "Accepted"
```

```
    review_status = "Not Required"
```

5. Calculate Delivery Charges

```
del_charge = 30.0 # Base delivery fee
```

```
if distance > 5:
    excess_distance = distance - 5
    del_charge = del_charge + (excess_distance * 10.0)
```

```
if weather == "Rainy":
    del_charge = del_charge + 20.0
elif weather == "Stormy":
    del_charge = del_charge + 50.0
```

```
if demand == "High" or is_peak_hour == True:
    del_charge = del_charge + 25.0
```

6. Calculate Applied Discount

```
discount = 0.0
if cust_type == "VIP":
    discount = order_amt * 0.15
elif order_amt > 1000:
    discount = order_amt * 0.10
elif order_amt > 500:
    discount = order_amt * 0.05
```

7. Determine Priority Status

```
if cust_type == "VIP" or (order_amt > 1500 and demand != "High"):
    priority_status = "High"
else:
    priority_status = "Standard"
```

8. Determine Final Order Category

```
if order_amt > 2000:
    order_category = "Premium"
```

else:

order_category = "Standard"

9. Calculate Final Payable Amount

final_payable = order_amt + del_charge - discount

if final_payable < 0:

final_payable = 0.0

10. Output Structured Summary Report

print("\n=====")

print(" FINAL ORDER REPORT ")

print("=====")

print("Order Status: ", order_status)

print("Restaurant Status: ", rest_status)

print("Cancellation Risk: ", cancel_risk)

print("Manual Review Status: ", review_status)

print("Priority Delivery Status:", priority_status)

print("Final Order Category: ", order_category)

print("-----")

print("Base Order Amount: ", round(order_amt, 2), "INR")

print("Delivery Charge: +", round(del_charge, 2), "INR")

print("Discount Applied: -", round(discount, 2), "INR")

print("Final Payable Amount: ", round(final_payable, 2), "INR")

print("=====")

11. Control Loop Continuation

keep_running = input("\nDo you want to process another order? (yes/no): ").strip().lower()

print("\nExiting system. Goodbye!")

Output:

```
--- Start a New Food Delivery Order ---
Enter order amount (INR): 435
Enter delivery distance (km): 5
Enter customer type (VIP/Regular): Vip
Enter customer rating (1.0 - 5.0): 4.5
Enter restaurant rating (1.0 - 5.0): 4
Enter preparation time (minutes): 20
Enter payment method (Prepaid/COD): Cod
Enter weather condition (Clear/Rainy/Stormy): Stormy
Enter demand level (Low/Normal/High): Normal
Is it a peak hour? (Yes/No): Yes
Enter number of previous cancellations: 0

=====
                FINAL ORDER REPORT
=====
Order Status:           Manual Review
Restaurant Status:      Reliable
Cancellation Risk:      Low
Manual Review Status:   Pending
Priority Delivery Status: High
Final Order Category:   Standard
-----
Base Order Amount:      435.0 INR
Delivery Charge:        + 105.0 INR
Discount Applied:       - 65.25 INR
Final Payable Amount:   474.75 INR
=====

Do you want to process another order? (yes/no): no

Exiting system. Goodbye!
```